

Different pedagogies for acquisition of knowledge and skill: a systematic review and network meta-analysis

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ABSTRACT

Background This network meta-analysis aimed to assess the efficacy of different educational methods for healthcare professionals.

Methods A systematic literature search was conducted to identify relevant randomised controlled trials. The standardised mean difference (SMD) with 95% CI was estimated using network meta-analysis for knowledge acquisition and skill performance, and ranked the effects of different educational methods using the surface under the cumulative ranking area (SCURA) technique.

Results Fifty-five randomised controlled trials including of 4292 participants were identified. Compared with no intervention, the results demonstrated all education methods achieved significant improvements in knowledge acquisition (SMD 1.73–2.66). Only education methods involving high fidelity virtual patient simulation reported significantly better skill performance (SMD 1.25–1.81). High fidelity virtual patient simulation plus self-directed learning was the most effective educational method both in terms of knowledge acquisition (SMD 2.66, 95% CI 1.4 to 4.12, SCURA 0.78) and skill performance (SMD 1.81, 95% CI 0.42 to 3.2, SCURA 0.89).

Conclusions Our study demonstrates all educational methods have positive effects on knowledge acquisition, but education methods involving high fidelity virtual patient simulation are better at improving skill performance than other methods.

INTRODUCTION

Clinical training is essential for undergraduate and postgraduate courses for healthcare students. However, healthcare education in the clinical environment is demanding and complex, requiring medical knowledge, clinical skills and teaching competencies.¹ Traditional education methods, such as didactic teaching methods (DTM) (including lectures, seminar, reading assignments), self-direct learning (SDL) (including problem-based learning, case-based discussion and self-study) and mannequin-based training, are important for current medical training courses. However, such educational methods rarely allow medical students to train in an interactive way, so knowledge acquisition and skills improvement may be less effective. With the development of modern technology, digital learning can enhance the learning experience of traditional teaching methods. Digital learning or e-learning includes mobile learning,

online and offline learning, virtual patient simulation (VP) and high-fidelity virtual patient simulation (HFVP).^{2,3} VP and HFVP may involve use of computerised mannequins of varying complexity to allow more realistic simulations of real-life clinical scenarios for the purpose of healthcare training and education.^{3,4} The trainees make clinical decisions to establish a correct diagnosis and learn optimal clinical management. In HFVP simulation, the mannequin is very realistic and enables accurate and interactive teaching.^{5,6} Blended or hybrid learning is a combination of traditional place-based classroom teaching with online educational materials.⁷

In previous studies, digital learning has shown promising improvements in medical knowledge and skills in the field of paediatrics,² more cost-effective antibiotic management,⁸ and was as effective as traditional methods of communication skills training in medical students.⁹ Blended learning has been found to be at least as effective as traditional educational methods.¹⁰ Virtual patients can more effectively improve skills, and at least as effectively improve knowledge.³ Previous meta-analyses have compared virtual patient simulations with traditional education or no intervention, revealing the improvement of knowledge and skill performance.^{2,3,8,10,11}

However, these meta-analyses were hard to interpret due to the absence of integrated multiple comparisons. Thus, this network meta-analysis evaluated the relative efficacy of different methods of education in healthcare to guide future improvements in the quality of training of healthcare students and professionals.

METHODS

This systematic review and network meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses for Network Meta-Analyses (see online supplemental appendix 1). The review protocol was prespecified and registered in the Open Science Framework (protocol available at <https://osf.io/x97y4>).

Search strategy

We performed a comprehensive search without language restrictions using electronic databases (including PubMed, Embase, the Cochrane Central Register of Controlled Trials, Cochrane Database of Systematic Reviews, China National Knowledge Infrastructure (<https://www.cnki.net>) and



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Web of Science) from inception to 1 July 2019, to identify all published and unpublished trials, with screening of the titles, and reviewing of the abstracts. We adapted the search strategy and keywords presented in online supplemental appendix 2 for the above-mentioned databases. To ensure no missing randomised controlled trials (RCTs), we hand searched the grey literature (conference abstracts and doctoral theses) and reference lists of included articles. For ongoing trials, we collected information from Google Scholar, the US government clinical trials database (www.ClinicalTrials.gov) and the UK National Research Register.

Study selection

We included parallel-group randomised studies, quasi-randomised control studies and cluster randomised trials in either postregistered or preregistered healthcare students and professionals reporting at least one outcome of interest (knowledge, skill), receiving virtual patient simulation (VP), HFVP simulation, SDL, digital education (DE), DTM, HFVP simulation plus SDL (HFVP +SDL) or HFVP simulation plus DTM (HFVP+DTM). The comparison must have included two or more variants of educational methods. Students from disciplines such as medicine, nursing, medical diagnostic and treatment technology, physiotherapy and rehabilitation, and pharmacy were enrolled in these studies. To prevent a carryover effect, we excluded crossover trials.

Two review authors (H-JJ, L-JO-Y) independently identified the trials meeting the eligibility criteria by screening the titles and abstracts, and another author (P-HC) adjudicated differences. In case of disagreement, another author (C-HL) was consulted, and a group discussion was organised to make the final decision.

Data extraction and bias assessment

Two reviewers (H-JJ and P-HC) independently assessed eligibility for all identified citations and extracted data. Data extraction was performed with a specifically designed form to capture information on study-related (author name, year of publication, study design and country), participant-related (sample size, participant's characteristics and measurement tools) and intervention-related (knowledge and skill) characteristics. For relevant data that were missing or not clearly reported, we contacted the corresponding authors for the required information. The same authors also used the Cochrane Risk of Bias (RoB) tool¹² independently to appraise selected studies. Seven domains were evaluated as high, unclear or low risk: these were sequence generation, allocation concealment, blinding of participant and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting bias and other bias. We produced RoB graphs using Review Manager V.5.3 software.¹³ If there were discrepancies in the data extraction or quality appraisal, we came to a consensus in consultation with a third reviewer (C-HL), or by deliberation within the group.

Outcome measures

Definition of knowledge

Knowledge was defined as delivery of the theoretical basis of clinical practice, measured by paper-based tests including multiple-choice questions, multiple-response questions, true/false questions, key feature format questions, free-text answers or short answer questions.

Definition of skill

Skill was defined as the student's ability to demonstrate clinical performance, assessed on mannequin, a live standardised patient

or virtual patients, measuring by written assignment, mix of paper cases and virtual patients, computer-based assessment or objective structured clinical examination.

Data synthesis and statistical analysis

We first conducted traditional pairwise meta-analyses for including studies to directly compare different educational methods. Both the fixed-effect and DerSimonian-Laird random-effects models were performed using the inverse-variance method, we calculated the ORs with the 95% CIs of direct comparisons between two methods. The heterogeneity was evaluated using the Cochran Q test and I^2 statistics. When the p value was <0.1 or the I^2 value was $>50\%$, heterogeneity was considered significant. Then, we conducted a network meta-analysis using a frequentist approach to compare different educational methods. This provided a point estimate from the network, and 95% CI from the frequency distribution of the estimate.

We examined the symmetry and geometry of the evidence by producing a network plot with nodes for the study subjects. The contribution of each direct comparison to the estimation of the network summary effects was calculated according to the combination of the variance of the direct treatment effect, and the network structure¹⁴ (see online supplemental appendix 3).

For skill and knowledge results, we calculated the standardised mean difference (SMD) with 95% CIs. A positive SMD value indicated a greater increase in both of the outcomes.

The probability of a mode being ranked was calculated as their surface under the cumulative ranking curve (SUCRA), which is the percentage of efficacy achieved by an approach compared with an imaginary approach that is always the best without uncertainty (ie, SUCRA 100%). SUCRA provides a hierarchy of treatments and accounts for variance of all relative intervention effects.^{15–17} The SUCRA value is 0 when a treatment is the worst option, and 1 when a treatment is the best option. Higher SUCRA scores correspond to better knowledge acquisition and skill performance.¹⁸ Forest plots represented the overview of the results of the network meta-analysis and summary relative mean effects, 95% CI and SUCRA for all comparisons.¹⁹ We summarised values of SUCRA in the two-dimensional scatter plots, which represented the probability of being the best educational method in terms of both knowledge acquisition and skill performance.

We used a multivariate random-effects meta-regression with a consistency model, as described by White *et al.*²⁰ We assessed potential inconsistencies by comparing deviance, and deviance estimates, for each comparison between consistency and inconsistency using a random-effects design-by-treatment interaction model^{21 22} and a node-splitting technique.^{23 24} Statistical significance was considered as $p<0.05$ for both analyses.

Network transitivity was assessed by visually inspecting tables with study-related characteristics that may modify treatment effects, including difference in patient characteristics, study designs, details of the intervention and differences in outcome measurement.

We evaluated whether treatment effects for the two outcomes were robust in multiple meta-regression and subgroup analyses. The following effect modifiers were analysed by a multiple meta-regression model: fields of disciplines, study designs, educational status, study location and assessment measurement tools. Subgroup analyses were performed with the following variables: fields of disciplines (such as medicine or nursing), performing studies in east or west countries, and educational scenario (cardiopulmonary or non-cardiopulmonary).

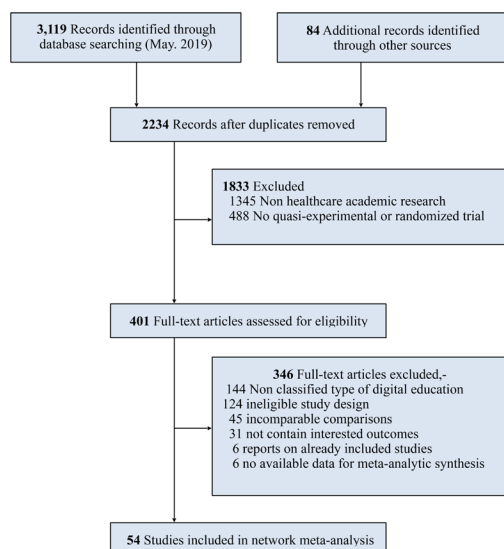


Figure 1 PRISMA flow diagram of network meta-analysis. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

The sensitivity analyses of our conclusions were evaluated by analysing the dataset with the following restrictions: studies without comparison of no intervention, studies without multiple arms design, and studies without quasi/cluster RCTs. Comparison-adjusted funnel plots were constructed, and the Begg's test²⁵ was applied to evaluate publication bias and small study effects.¹⁵

All network meta-analyses were performed with the statistical package 'netmeta' 0.9–0 (<https://cran.r-project.org/web/packages/netmeta/index.html>) in R V.3.4.2 (R Core Team, Vienna, Austria) and Stata, V.14 (StataCorp).

Quality assessment

The quality of evidence derived from the network meta-analysis was judged by use of the Grading of Recommendations Assessment, Development and Evaluation (GRADE) criteria.^{26–27} First, the quality of direct and indirect evidence was assessed separately. If randomised trials formed the evidence, the direct evidence rating started at high quality, but could be downrated by any concerns about the RoB, inconsistency, indirectness, imprecision or publication bias. The rating of indirect evidence began with the lower evidence rating of the two pairwise direct comparisons of first order loops, but could be downrated further for imprecision, or intransitivity (differences between studies that form the basis for the indirect evidence in clinical or methodological characteristics). Thereafter, when both direct and indirect evidence were available, the higher of the two quality ratings was suggested by the GRADE Working Group as the preferable rating of the quality of effect estimates in a network meta-analysis.

RESULTS

Systematic literature review

The initial screening retrieved 3119 article titles and abstracts from the electronic databases. After removing duplicates, 2234 citations remained. Further screening using the title or abstract to meet the clinical trial requirement excluded 1833 studies. In the next phase, 401 full texts were assessed for potential eligibility, and 346 studies were excluded for a variety of reasons. The remaining 54 studies (49 two-arm studies, 3 three-arm studies and 2 four-arm study) were high quality, reported as

complete research articles, and all were included in our quantitative synthesis (figure 1).

The 54 trials (19 quasi-RCT, 35 RCT) investigated a total of 4292 participants randomised to the following 8 educational methods: virtual patient simulation (VP), HFVP simulation, SDL, DE, DTM, HFVP +SDL, HFVP +DTM and no intervention.

The participants mostly came from the fields of medicine (47.3%) and nursing (49.1%), including both undergraduates (89.1%) and postgraduates (10.9%). Trials were conducted in North America (43.6%), Europe (29.1%) and Asia (18.2%). The training scenario evaluated included cardiology, pulmonology, gastroenterology, otolaryngology, psychiatry, ophthalmology, paediatrics and acute care. Thirteen studies evaluated both the knowledge and skill improvement of the participants after receiving different teaching methods, 34 studies evaluated knowledge improvement, and 34 studies evaluated skill improvement (see online supplemental appendix 4).

Results of knowledge assessment

In terms of knowledge assessment, 34 studies were included, with a total of 2976 participants and 8 educational options (figure 2A). All education options, including 'no intervention', were compared with at least two other education options in any of the networks. In the comparison of simulation education, only HFVP+SDL was not directly compared with 'no intervention'.

Transitivity was maintained across comparisons of interventions for the outcomes of knowledge assessment. We found that all education methods were more effective than placebo, with SMD ranging between 2.66 (95% CI 1.20 to 4.12) for HFVP+SDL, and 1.73 (95% CI 0.66 to 2.80) for DTM (figure 3A). The SUCRA value of HFVP+SDL was 0.78, showing that the cumulative ranking probability of being the best treatment option in the network analysis was 78%, the highest across all the interventions (figure 3A).

Results of skill assessment

In terms of skill performance, 34 studies were included, with a total of 2399 participants and eight education options (figure 2B). All education options, except HFVP+SDL, were compared with at least two other education options in any of the networks. Transitivity was maintained across intervention comparison for the outcomes of skill performance. All interventions were sorted based on their ranking and accompanied by the SMD with 95% CI vs the comparator 'no intervention'. Education methods involving HFVP simulation were significantly more effective than no intervention, with SMD ranging between 1.81 (95% CI 0.42 to 3.21) for HFVP+SDL and 0.97 (95% CI: 0.02 to 1.92) for DE (figure 3B). The SUCRA value of HFVP+SDL at 0.89 was the highest across all the interventions (figure 3B).

The detailed results of the network meta-analysis are presented in figure 4. Figure 5 presents the two-dimensional graphs about improvement of knowledge and skill in all studies. HFVP+SDL was most the most efficacious education method in terms of both knowledge acquisition and skills performance, while HFVP +DTM and HFVP alone were ranked second equal.

Inconsistency, RoB and publication bias

We evaluated the inconsistency of these outcomes using the design-by-treatment interaction model, and the node-splitting model (see online supplemental appendix 5). The inconsistency analyses showed that there may be some inconsistency between 'no intervention' versus other educational methods (see online

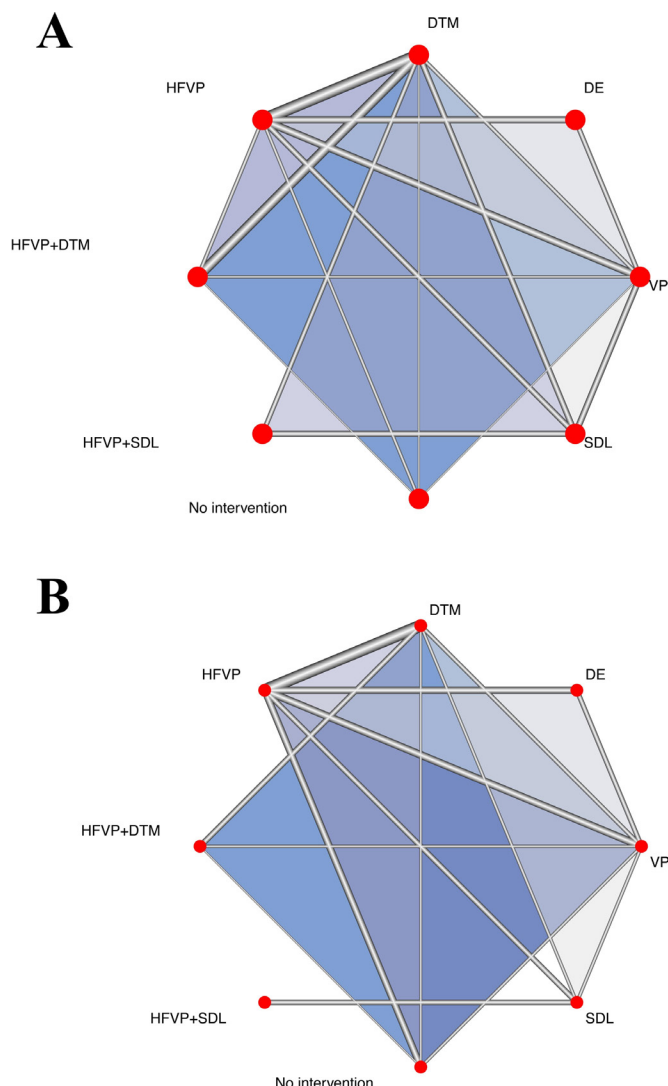


Figure 2 Schematic diagram of the network of evidence used in network meta-analysis. (A, B) is arranged in order of major outcomes: (A) knowledge and (B) skill. DE, digital education; DTM, didactic teaching methods; HFVP, high-fidelity virtual patient simulation; SDL, self-directed learning; VP, virtual patient simulation.

supplemental appendix 6). We tried to reveal the potential inconsistency from both clinical and statistical points of view. We used a random-effect multiple meta-regression model with variables of medical field, study design, academic stage, location and assessment method for the outcomes of skill performance and knowledge acquisition (see online supplemental appendix 7). The results showed the different medical fields had an impact on the performance of skills when using different educational methods (see online supplemental appendix 7.1.1). We also conducted sensitivity analyses of omission of multiple arm studies, excluding ‘no intervention,’ and excluding multiple arm trials (see online supplemental appendix 8). These showed diminished inconsistency of both the outcomes if ‘no intervention,’ or studies with multiple arms were excluded (see online supplemental appendix 8). Subgroup analyses by medicine field, nurses field and publication location were also carried out for potential bias evaluation (see online supplemental appendix 8). The RoB assessment for enrolled RCTs is illustrated and summarised in the appendix 9. High RoB was rare in any domain. However, unclear assessments were common, because some articles did not

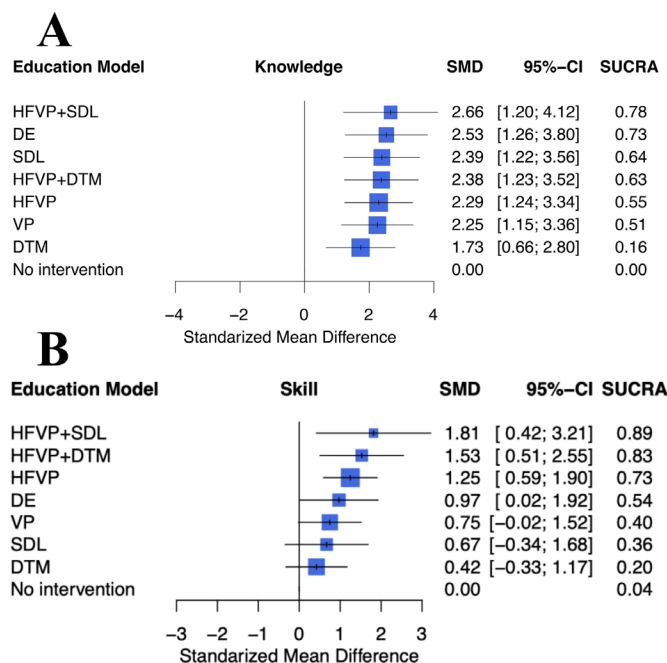


Figure 3 Network meta-analysis results of knowledge and skill. (A, B) is arranged in order of major outcomes: (A) knowledge and (B) skill. DE, digital education; DTM, didactic teaching methods; HFVP, high-fidelity virtual patient simulation; SDL, self-directed learning; SMD, standardised mean difference; SUCRA, surface under the cumulative ranking curve; VP, virtual patient simulation.

Outcome: Knowledge (Measure: Standardized mean difference)									
HFVP+SDL	-	0.29 (-0.74 to 1.31)	-	-	-	0.92 (-0.51 to 2.35)	-	-	-
0.13 (-1.14 to 1.40)	DE	-	-	0.39 (-0.49 to 1.26)	-0.08 (-1.13 to 0.97)	-	-	-	-
0.27 (-0.70 to 1.24)	0.14 (-0.77 to 1.05)	SDL	-	-0.04 (-0.90 to 0.81)	0.17 (-0.69 to 1.03)	0.81 (-0.22 to 1.84)	-	-	-
0.29 (-0.88 to 1.45)	0.15 (-0.81 to 1.12)	0.01 (-0.78 to 0.81)	HFVP+DTM	-0.89 (-2.35 to 0.57)	-0.09 (-1.58 to 1.41)	0.67 (0.11 to 1.23)	4.03 (2.33 to 5.74)	-	-
0.37 (-0.68 to 1.42)	0.24 (-0.51 to 0.99)	0.10 (-0.48 to 0.68)	0.08 (-0.56 to 0.73)	HFVP	0.35 (-0.26 to 0.97)	0.51 (0.01 to 1.00)	0.61 (-0.87 to 2.08)	-	-
0.41 (-0.68 to 1.50)	0.28 (-0.50 to 1.06)	0.14 (-0.48 to 0.76)	0.12 (-0.61 to 0.85)	0.04 (-0.44 to 0.51)	VP	1.16 (0.09 to 2.22)	4.12 (2.40 to 5.84)	-	-
0.93 (-0.12 to 1.98)	0.80 (-0.04 to 1.64)	0.66 (0.03 to 1.29)	0.64 (0.10 to 1.18)	0.56 (0.13 to 0.99)	0.52 (-0.05 to 1.09)	DTM	3.12 (1.49 to 4.75)	-	-
2.66 (1.20 to 4.12)	2.53 (1.26 to 3.80)	2.39 (1.22 to 3.56)	2.38 (1.23 to 3.52)	2.29 (1.24 to 3.34)	2.25 (1.15 to 3.36)	1.73 (0.66 to 2.80)	No intervention	-	-
Legend: Network Meta-analysis Estimate (green), Comparator (blue), Direct Estimate (orange)									

Outcome: Skill (Measure: Standardized mean difference)									
HFVP+SDL	-	-	-	-	1.15 (0.18 to 2.11)	-	-	-	-
0.29 (-1.21 to 1.79)	HFVP+DTM	-	-	-0.02 (-1.65 to 1.60)	-	0.97 (0.15 to 1.80)	3.58 (1.80 to 5.36)	-	-
0.57 (-0.68 to 1.81)	0.28 (-0.60 to 1.16)	HFVP	0.11 (-0.70 to 0.91)	0.84 (0.22 to 1.47)	0.65 (-0.30 to 1.60)	0.89 (0.35 to 1.44)	0.49 (-0.25 to 1.24)	-	-
0.84 (-0.58 to 2.27)	0.56 (-0.56 to 1.67)	0.28 (-0.44 to 0.99)	DE	0.08 (-1.07 to 1.23)	-	-	-	-	-
1.07 (-0.24 to 2.38)	0.78 (-0.17 to 1.73)	0.50 (-0.01 to 1.01)	0.22 (-0.56 to 1.01)	VP	0.35 (-1.26 to 1.96)	0.93 (-0.21 to 2.08)	2.01 (0.79 to 3.22)	-	-
1.15 (0.18 to 2.11)	0.86 (-0.29 to 2.01)	0.58 (-0.21 to 1.37)	0.30 (-0.75 to 1.36)	0.08 (-0.81 to 0.97)	SDL	0.82 (-0.79 to 2.43)	-	-	-
1.39 (0.09 to 2.70)	1.11 (0.32 to 1.90)	0.83 (0.35 to 1.31)	0.55 (-0.29 to 1.40)	0.33 (-0.31 to 0.97)	0.25 (-0.63 to 1.13)	DTM	2.15 (0.87 to 3.43)	-	-
1.81 (0.42 to 3.21)	1.53 (0.51 to 2.55)	1.25 (0.59 to 1.90)	0.97 (0.02 to 1.92)	0.75 (-0.02 to 1.52)	0.67 (-0.34 to 1.68)	0.42 (-0.33 to 1.17)	No intervention	-	-
Legend: Network Meta-analysis Estimate (green), Comparator (blue), Direct Estimate (orange)									

Figure 4 Results of the head-to-head comparison of knowledge and skill assessment in network meta-analysis. Figure 4A,B is arranged in order of major outcomes: (A, knowledge and B, skill). Data are presented as the standard mean difference with 95% CI in the column-defining intervention compared with the row-defining intervention. Comparisons should be read from left to right. DE, digital education; DTM, didactic teaching methods; HFVP, high-fidelity virtual patient simulation; SDL, self-directed learning; VP, virtual patient simulation.

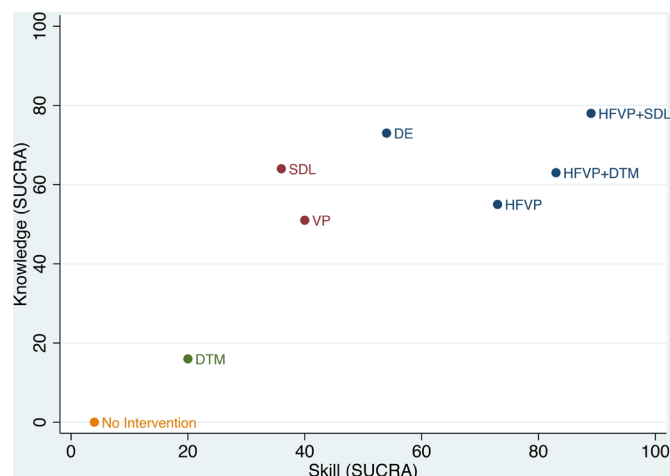


Figure 5 Scatter plot present the results of our network meta-analysis. The closer to the right top half means the intervention has better improvement of knowledge and skill. X-axis represents SUCRA value of skill; Y-axis presents SUCRA value of knowledge. Both of the values were showed in brackets. Navy nodes indicate the most effective treatment options. Red nodes indicate treatment options more effective than no intervention. Green blue nodes indicate treatment options has mild improvement comparing to NO intervention. DE, digital education; DTM, didactic teaching methods; HFVP, high-fidelity virtual patient simulation; SDL, self-directed learning; SUCRA, surface under the cumulative ranking area; VP, virtual patient simulation.

describe the randomisation and allocation methods. Regarding selective reporting, trials judged as having low RoB were 92% and 85% for knowledge acquisition and skills performance outcomes, respectively (see online supplemental appendix 9).

Online supplemental appendix 10 shows the comparison-adjusted funnel plots of knowledge and skill outcomes, and these revealed no significant visual asymmetry. Lastly, the Begg's test revealed no evidence of small study bias.

The GRADE results and network comparison of knowledge and skill outcomes are shown in online supplemental appendix 11.

DISCUSSION

The strength of our systematic review is the broad perspective on RCTs involving healthcare educational methods. Our systematic review also summarises the evidence on effectiveness of these methods using the GRADE system. Our results showed that all education methods produced significant improvements in knowledge acquisition, with VP and HFVP methods having similar effects. In terms of skill performance, we found that education methods involving HFVP simulation were better than traditional methods, with HFVP, and HFVP +SDL or HFVP +DTM being the most effective. This phenomenon was previously described by Kononowicz *et al*,³ and relates to the lower levels of Bloom's taxonomy.²⁸ Knowledge transmission, understanding and memory processes are all time-dependent so one would expect that all education methods would have some effect over 'no intervention'.

Lots of educational methods are developed to enhance the performance of clinical staff; nevertheless, this might result in the burden of teachers to get familiar with these education methods and encounter a pedagogies choosing dilemma. Therefore, it is a crucial unmet need to assist teachers to identify the most appropriate one. In the network analyses,

we estimate the effect size using the SMD method for the outcomes measured in different ways and provide the relative effect size of each method by ranking. The relative ranking in different teaching scenarios provides a general comparison on how to choose a better teaching modality. Based on the concept, teachers could adjust teaching strategy to meet the student's learning needs. Our results denote that the education method involving HFVP had the better performance in knowledge and skill.

HFVP provides close-to-reality experiences, and includes features such as realistic physiological responses, communication and interaction with the mannequin. Practical 'hands-on' applications allow practice of skills and tasks. Where knowledge acquisition is combined with skills performance and applied to clinical problem solving, HFVP is particularly effective, especially when direct patient contact is not yet possible. Our network meta-analyses provides strong evidence that HFVP is the best educational method for skill and knowledge training.

Although there are guidelines for education of healthcare students and professionals, the tutor's personal characteristics, the protocol of the education method, the duration of the learning period and even the assessment tools all vary.^{29–32} The results of our meta-analysis were affected by a potential inconsistency in that there was a significant relationship between the different healthcare fields and the efficacy of different educational methods, but this will need a larger study to explore further. We also found some bias may have existed when educational methods were compared with 'no intervention,' so we recommend that future studies should directly compare two educational methods. Unfortunately, most studies have low statistical power, and it is difficult to define the most suitable populations to study. Our meta-analysis should help design further studies as a basis for evidence-based guidelines, protocols and algorithms.

Cost-effectiveness was not the focus of our study but is an important issue. The cost ingredients included equipment (simulator, audiovisual for recording and viewing), training materials, cost of personnel (technicians, administrative staff), facility costs, learner costs (transportation, meals, course registration) and opportunity costs (lost clinical revenue from taking time to learn or teach). As an example, Isaranuwachai *et al* conducted a cost-effectiveness analysis of simulation modalities for peripheral intravenous catheterisation training. Fifteen medical students per model were enrolled, and the total cost of a low fidelity VP method was US\$16 300, compared with US\$67 727 for a high fidelity VP method.³³ Zendejas *et al* found the cost of mannequin-based simulation training ranged from US\$160 to US\$800 per hour.³⁴

Results of the study should be elucidated in light of both strengths and limitations. To assess the same outcome but measure with different ways, we standardised the results of the studies to a uniform scale using SMD and estimated relative rankings of education interventions to quantify the influence of a study and provide the robust evidence for clinical instructor. We aggregated eight educational methods and the number of participants in individual studies was relatively small, which did not allow nuances in the studies to be explored thoroughly. Although we included 55 trials, the evidence to determine the effective factors was sparse, with very diverse research questions.

In conclusion, our network meta-analysis provides the evidence that all educational methods have positive effects on knowledge acquisition, but education methods involving HFVP simulation are better at improving skill performance than other methods.

Main messages

- ⇒ Novel digital education methods involving high fidelity virtual patient simulation have been developed for training healthcare students and professionals to improve their clinical problem solving abilities.
- ⇒ For knowledge acquisition, all educational methods have positive effects on knowledge acquisition.
- ⇒ For skill performance, high fidelity virtual patient simulation plus self-directed learning appeared to be most effective education method.

Current research questions

- ⇒ Have novel digital education methods improved training healthcare students and professionals abilities to solve clinical problem?
- ⇒ Based on network meta-analysis, which education methods is most effective?
- ⇒ What is the advantage of novel digital education methods?

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